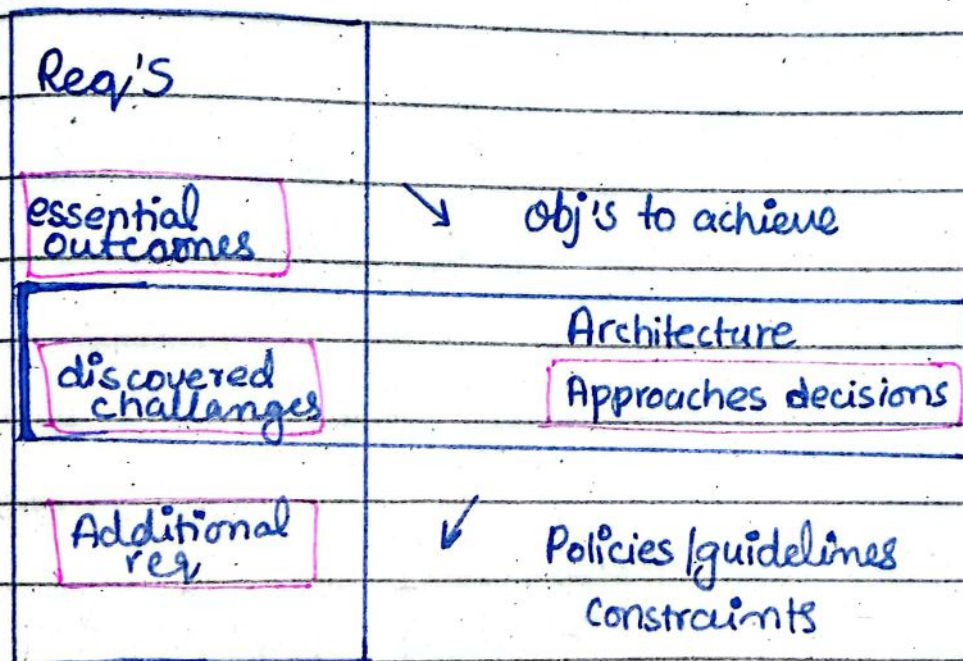


Interaction b/w requirement & architecture:-

- Requirements and req't architecture are closely intertwined in system engineering. Architecture represents the conceptual structure and behaviour of system, while requirements specify what the system should do.



Requirement as input to architecture

- Functional requirement:-

Functional req specify what the system should do in terms of behaviour and functionality. These requirements serves as input to the architectural design process, guiding the selection of system components, interfaces and interactions.

• Non-Functional requirement:

These requirements influence architectural decisions by defining constraints and quality goals that the architecture must satisfy.

Architecture as enabler of requirements:

• decomposition of requirements:

The architecture decomposes high-level requirements into lower level requirements components, modules and interfaces. It defines how system functionality will be implemented to fulfill specific requirements.

• Allocation of Responsibilities:-

architecture allocates responsibilities to different system components, defining their role and interactions, it ensures each req. is addressed by one or more architecture elements

• Trade off and Design choices:

Architecture decisions involve trade off's and design choices that can impact the feasibility and implementation of req.

" Architecture help ensures that set of requirement is complete "

Architecture design is critical link b/w requirement and design.

Similarities between Interaction and Architecture:

Aspect	Interaction	Architecture
System Understanding	Both interaction and architecture contribute to a deeper understanding of the system's behavior, functionality, and dependencies.	Both interaction and architecture help stakeholders visualize and conceptualize the system's structure, behavior, and relationships.
Collaboration	Both interaction and architecture involve collaboration between stakeholders, including developers, designers, architects, and end-users.	Both interaction and architecture require alignment with stakeholders' needs, requirements, and objectives.
Iterative Refinement	Both interaction and architecture may undergo iterative refinement throughout the software development lifecycle.	Both interaction and architecture may evolve in response to changing requirements, feedback, or insights gained during development.
Communication	Both interaction and architecture rely on effective communication to convey system design decisions, constraints, and requirements.	Both interaction and architecture facilitate communication between stakeholders, fostering shared understanding and consensus.